

KUUSAMO, Finland

WMO: 028690

Lat: 65.996N

Lon: 29.223E

Elev: 264

StdP: 98.19

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.4	-28.0	-34.6	0.1	-31.3	-30.9	0.2	-27.9	8.7	-5.0	7.8	-4.9	0.7	80	0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.3	(c) 24.5	(d) 16.2	(e) 22.4	(f) 15.3	(g) 20.6	(h) 14.6	(i) 17.8	(j) 21.9	(k) 16.6	(l) 20.5	(m) 15.5	(n) 19.3	(o) 3.6	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	11.9	19.6	14.9	11.0	18.2	13.8	10.2	17.2	51.2	21.7	47.4	20.5	44.2	19.0	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 7.8	(b) 6.9	(c) 6.2	(e) -34.7	(f) 26.9	(g) 3.6	(h) 2.0	(i) -37.3	(j) 28.3	(k) -39.4	(l) 29.5	(m) -41.4	(n) 30.7	(o) -44.0	(p) 32.1		
			DB													
			WB													
				-34.7	19.2	3.5	1.4	-37.3	20.2	-39.3	21.0	-41.3	21.8	-43.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.5	-12.1	-11.6	-7.5	-1.1	5.2	11.4	14.9	12.3	7.2	0.9	-5.2	-9.2
	DBStd	10.69	7.83	7.35	5.63	4.54	4.66	3.94	3.45	3.34	3.55	4.21	5.77	7.31
	HDD10.0	3806	684	604	543	332	161	29	4	16	97	283	457	596
	HDD18.3	6521	942	837	801	582	406	211	116	190	334	541	707	854
	CDD10.0	338	0	0	0	0	13	71	155	86	13	0	0	0
	CDD18.3	13	0	0	0	0	0	2	9	2	0	0	0	0
	CDH23.3	109	0	0	0	0	2	28	62	17	0	0	0	0
	CDH26.7	10	0	0	0	0	0	3	6	2	0	0	0	0
Wind	WSAvg	2.8	2.6	2.8	2.9	2.8	3.0	3.0	2.7	2.6	2.9	2.9	2.8	2.9
Precipitation	PrecAvg	598	40	32	31	34	47	62	76	71	58	52	50	41
	PrecMax	791	88	73	58	69	98	148	150	141	117	114	111	93
	PrecMin	416	9	5	7	4	2	11	10	25	9	15	8	12
	PrecStd	87	17	16	13	17	21	29	33	27	25	25	22	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.8	2.8	5.6	13.4	23.1	26.5	27.3	25.5	18.7	10.9	6.4	2.9
		MCWB	1.1	1.7	2.2	7.2	12.7	16.1	17.8	17.8	13.6	9.7	5.7	2.3
	2%	DB	0.2	0.9	3.1	9.7	19.5	23.1	25.1	22.4	16.0	9.1	4.3	1.4
		MCWB	-0.1	0.4	0.8	5.0	11.8	14.8	16.7	15.9	12.2	8.1	3.8	0.9
	5%	DB	-0.9	-0.3	1.4	7.2	16.1	20.6	23.1	20.2	14.2	7.8	2.5	0.5
		MCWB	-1.2	-0.7	-0.1	3.4	9.9	13.5	16.0	15.0	11.3	6.9	2.1	0.2
	10%	DB	-2.2	-1.8	0.1	5.5	13.3	18.5	21.2	18.2	12.8	6.4	1.2	-0.4
		MCWB	-2.5	-2.3	-0.9	2.4	8.2	12.4	15.3	14.1	10.8	5.7	0.8	-0.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.1	1.7	2.7	7.5	14.2	18.0	19.7	18.5	14.5	9.8	5.9	2.3
		MCDB	1.5	2.4	5.0	11.8	19.7	22.7	24.5	23.5	17.3	10.4	6.2	2.5
	2%	WB	0.0	0.3	1.2	5.7	12.4	16.2	18.4	17.1	13.0	8.3	3.9	1.0
		MCDB	0.2	0.8	2.5	8.7	18.9	21.2	22.5	21.0	15.0	8.9	4.3	1.3
	5%	WB	-1.1	-0.8	0.2	3.9	10.5	14.7	17.2	15.7	12.0	7.0	2.1	0.2
		MCDB	-0.9	-0.4	1.1	6.5	15.7	19.7	21.2	18.8	13.8	7.6	2.4	0.4
	10%	WB	-2.5	-2.3	-0.8	2.6	8.6	13.2	16.1	14.6	11.0	5.7	0.9	-0.8
		MCDB	-2.3	-1.9	0.1	4.8	12.4	17.0	19.9	17.5	12.5	6.3	1.1	-0.5
Mean Daily Temperature Range	5% DB	MDBR	7.3	7.7	9.5	8.8	8.5	8.5	8.3	8.0	6.6	4.3	5.2	6.4
		MCDBR	6.0	5.5	8.8	11.0	13.7	12.1	11.5	11.2	9.0	5.2	4.1	4.7
		MCWBR	6.0	5.3	7.2	7.1	7.3	5.5	4.8	5.6	5.7	4.4	4.1	4.6
	5% WB	MCDBR	6.0	5.4	7.1	9.7	12.6	10.6	9.2	9.1	7.7	4.6	4.0	4.6
		MCWBR	6.0	5.2	6.1	6.4	7.2	5.5	4.8	5.3	5.3	4.3	4.0	4.6
Clear-Sky Solar Irradiance	taub	0.194	0.262	0.295	0.327	0.339	0.339	0.357	0.336	0.312	0.284	0.351	0.194	
	taud	1.926	2.107	2.160	2.247	2.453	2.489	2.458	2.491	2.540	2.363	2.700	1.950	
	Ebn at Noon	245	594	761	833	865	873	842	826	769	588	256	35	
	Edh at Noon	18	55	86	103	94	93	93	81	62	43	18	4	
All-Sky Solar Radiation	RadAvg	0.09	0.59	2.04	3.81	4.73	5.06	4.73	3.44	1.85	0.62	0.14	0.01	
	RadStd	0.01	0.11	0.19	0.37	0.44	0.38	0.47	0.34	0.19	0.10	0.02	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)		+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page